Toward the Computational and Ethical Foundations of Participatory and Adaptive AI Intervention Systems for Health Behavior Change

Health behaviors such as vaping, poor diet, physical inactivity, and medication nonadherence are increasingly shaped within digital environments where individuals encounter peer influence, rapidly evolving information, and algorithmically curated content. Short-form video and social media platforms are especially influential among young adults, creating important opportunities to deliver timely, personalized, and scalable health interventions. However, current digital health tools are often static, opaque, and designed *for* users rather than *with* them. Existing AI-driven systems typically optimize engagement rather than wellbeing, while overlooking users' concerns regarding privacy, transparency, autonomy, and fairness. As a result, many digital interventions fail to sustain trust, relevance, or long-term behavior change.

This CAREER project seeks to establish the **computational and ethical foundations of participatory and adaptive AI intervention systems for health behavior change**. The central premise is that effective intervention systems must not only infer users' behavioral needs and adapt support in real time, but also enable individuals to actively shape intervention content, personalization logic, and data governance. Rather than treating people solely as passive targets of personalization, the project reimagines users as active collaborators in the design of intelligent health support systems.

Grounded in Self-Determination Theory, the project will design intervention systems that support users' **autonomy** (control, consent, transparency), **competence** (timely and effective support), and **relatedness** (socially resonant communication). The project uses vaping prevention and cessation among young adults as a motivating domain, while developing methods intended to generalize to broader health behavior change applications.

Aim 1: Develop AI models of behavioral context, receptivity, and support needs

Create multimodal AI methods that infer readiness for change, barriers, patient-reported outcomes, and moments when users are most receptive to intervention. These models will integrate forum text, social media content, and interaction signals to support real-time adaptive decision making.

Aim 2: Create participatory AI systems for co-designing interventions

Develop human-AI systems that enable users to collaboratively create and refine intervention messages, short-form videos, delivery preferences, and personalization settings. This aim will study how co-design improves relevance, trust, and user ownership.

Aim 3: Design ethics-aware adaptive intervention systems

Build adaptive algorithms that jointly optimize behavioral effectiveness, privacy preferences, transparency, fairness, and sustained engagement. Controlled experiments on the prototype platform will compare participatory versus non-participatory systems and adaptive versus static interventions.

Intellectual Merit

This project advances human-centered computing by integrating AI, behavioral science, participatory design, and responsible technology development. It will produce new multimodal models of health-relevant user states, computational frameworks for AI-assisted co-design, and adaptive algorithms that explicitly incorporate ethical constraints. More broadly, it introduces a new paradigm in which intelligent intervention systems are designed collaboratively with users rather than imposed upon them.

Broader Impacts

The project will support healthier digital ecosystems through scalable intervention technologies and evidence-based design principles. It will create interdisciplinary training opportunities for students in AI, ethics, and digital health; develop curricular modules on responsible AI intervention systems; conduct participatory workshops with young adults and community partners; and release open-source tools and datasets to the broader research community. Findings will inform public health practitioners and technology designers seeking to build trustworthy AI systems for behavior change.